

The source sign of φ_{bb}

The split is exact; uniform negativity is refuted; the defect pole is not dominant

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Abstract

Round 417 reduced cofinal $\text{sch} < 0$ to $\varphi_{bb} < 0$ plus $|\tau_{\text{un}}| \rightarrow 0$. On the vacuous chart $\varphi_{bb} < 0$ makes the whole disk live; the scalar itself was never attacked. The closed split $\varphi_{bb} = c_J + s^T A_0^{-1} s$ is a finite SATZ, and the self-energy is the r407 C -resolvent of the Sherman–Morrison border. Uniform $\varphi_{bb} < 0$ on the named living census is **refuted** (six vacuous windows overflow). Pole-dominance — the hope that the single negative bulk mode is the source of edge negativity — is **refuted** on living P1 (3/28). Two named mechanisms. No RH claim.

Status. Lemma 1 is **refuted** as a world-blind law on the named census (and reduced to two mechanisms on the branches; selected $a_k = 2^k$ remains a negative census, not a theorem). The split and the C -resolvent form are finite identities (Proposition 1). Ausgang `SPLIT_EXACT` / `UNIFORM_REFUTED` / `POLE_NOT_DOMINANT`. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The scalar

Write $A_0 = R_{N-3} - \frac{1}{2}I$ on the hole nodes Y and $s = u_B$ the Sherman–Morrison border of r362/r401, with border scale $\text{den} > 0$. In the r369 Haynsworth matrix $\Phi_N = J^{-1} + U^T A_0^{-1} U$ the border-border entry is

$$\varphi_{bb} = [J^{-1}]_{bb} + s^T A_0^{-1} s = c_J + \Sigma, \quad c_J = \text{den} - 2, \quad (1)$$

where $[J^{-1}]_{bb} = \text{den}$ and the -2 is the last-slot inverse $(A^\dagger)_{\text{last}}^{-1} = (-1/2)^{-1}$. The r407 dictionary $R = C(I + C)^{-1}$ gives $A_0 = \frac{1}{2}(C - I)(C + I)^{-1}$ and therefore

$$A_0^{-1} = 2(C + I)(C - I)^{-1}, \quad \Sigma = s^T A_0^{-1} s. \quad (2)$$

Spectrally $\Sigma = \sum_i |\langle s, \psi_i \rangle|^2 / \lambda_i(A_0)$. On living P1 there is one negative bulk eigenvalue (the window-emergent defect). Split $D = |\langle s, \psi_- \rangle|^2 / \lambda_-$ (negative, so it *helps* $\varphi_{bb} < 0$) against the positive rest R_+ .

2 The lemma

Lemma 1 (φ_{bb} sign). $\varphi_{bb} < 0$ for every window of the selected sequence, source-pure: the border self-energy through the bulk propagator undershoots the explicit J -bound.

Verdict: REFUTED as a world-blind law on the named living census (six vacuous core windows have $\varphi_{bb} > 0$). On the computed selected anchors $a_k = 2^k$ ($k = 4, 5, 6, 7, 9$) the inequality holds as a census, not a theorem (no k_0). The identity (1)–(2) is not the lemma; it is the exact decomposition against which the sign was tested.

Proposition 1 (split and C -resolvent). *Over $\mathbb{Q}$, $\varphi_{bb} = c_J + \Sigma$ with $\Sigma = s^T A_0^{-1} s$, and $A_0^{-1} = 2(C + I)(C - I)^{-1}$. P1 toy $A_0 = \text{diag}(-\frac{1}{2}, 2, 3)$, $s = e_0$, $\text{den} = 1$: $c_J = -1$, $D = -2$, $R_+ = 0$, $\varphi_{bb} = -3$. Vacuous toy $A_0 = \text{diag}(2, 3, 4)$, $s = e_0$, $\text{den} = \frac{1}{2}$: $c_J = -\frac{3}{2}$, $\Sigma = \frac{1}{2}$, $\varphi_{bb} = -1$. The 2×2 diagonal C -toy matches (2) exactly. A pole-save toy $A_0 = \text{diag}(-\frac{1}{10}, \frac{1}{5})$, $s = (1, 1)$ has $\varphi_{bb} = -6$ and $\varphi_{bb}|_{D=0} = +4$ (the mutant construction is live).*

3 Two mechanisms

Proposition 2 (flagship and core). *At $k_z = 9$, $\varphi_{bb} = -0.08562$, $c_J = -0.39889$, $\Sigma = 0.31326$, $D = -0.01465$, $R_+ = 0.32792$. Removing the defect leaves $\varphi_{bb}|_{D=0} = -0.07097 < 0$: the flagship is not pole-saved. The C -resolvent residual is 10^{-12} . Dropping the -2 in c_J produces $+1.914$ and kills the sign.*

Proposition 3 (named census). *MAIN 28/28 P1 have $\varphi_{bb} < 0$; MAIN vacuous 8/14 have $\varphi_{bb} < 0$ and 6/14 overflow at $k_z \in \{12, 13, 16, 19, 39, 49\}$. $c_J \in [-0.540, -0.348]$ is negative on every core window; $\Sigma > 0$ on every vacuous window ($C \succ I$ forces $(C - I)^{-1} \succ 0$). Defect-saves ($\varphi_{bb} < 0$ but $\varphi_{bb}|_{D=0} \geq 0$): exactly three P1 windows ($k_z = 21, 29, 36$). Median $|D|/|\varphi_{bb}| = 0.076$ — not pole-dominant. EXT six rows, including $k_z = 119$ at $N = 1119$, all $\varphi_{bb} < 0$. Selected $a_k = 2^k$: $k = 5$ ($k_z = 17$) $\varphi_{bb} = -0.105$; $k = 9$ ($k_z = 116$, $N = 1433$) $\varphi_{bb} = -0.038$. $\log |\varphi_{bb}| \sim \log N$ slope -0.09 (flat of order 0.05).*

P1 mechanism. $c_J + R_+$ is already negative on 25/28 living P1 windows. The single negative bulk mode is a small correction, not the source of edge negativity. Pole-dominance — bulk defect and edge sign as one mechanism — is refuted on the living P1 branch.

Vacuous mechanism. $\Sigma > 0$ always. $\varphi_{bb} < 0$ if and only if $c_J < -\Sigma$. On six named small- N windows the self-energy overflows the J -bound.

4 Death is not a φ_{bb} overflow

Proposition 4 (chi worlds). *Dead χ_3 3/5 keep $\varphi_{bb} < 0$ (the three P1: 15, 19, 39); 2/5 have $\varphi_{bb} > 0$ (vacuous 23, 33). Dead χ_{3-15} is pole-dominated ($\varphi_{bb} = -1.018$, $D = -1.217$) and dead ($\text{sch} > 0$): the one large pole sits on a death window, not a living P1. Dead χ_{4-20} (vacuous) has $\varphi_{bb} = +0.054$. Live χ_3 34/37 have $\varphi_{bb} < 0$. Hence φ_{bb} is not the sole life/death carrier: dead P1 dies by the τ -terms of r417 while φ_{bb} stays negative.*

5 Kills

Dropping the border column leaves only K_2 ($\text{ind}_- K_2 = \text{ind}_- A_0$; no φ_{bb} scalar). A false c_J (forget the -2) flips the w9 sign. The defect-mutant does *not* flip w9 and *does* flip the three named P1 windows and the rational pole-save toy — the mutant tests the mechanics and confirms they are not pole-dominant on living P1. Scramble $\text{nneg}(A_0) = 21$; two-period $S = 21$ has $\text{nneg} = 4$. The r377 product $h_N h_{N+1}$ at w9 stays negative and is independent of the φ_{bb} sign.

6 What is a theorem, what is a census

Proposition 1 is a theorem (finite identities over $\mathbb{Q}$, machine-checked on the named census via the C -resolvent). Lemma 1 is refuted as a uniform source law: six vacuous living windows have $\varphi_{bb} > 0$, so φ_{bb} is not a world-blind sign carrier. On P1 the J -bound wins against a positive rest; on the vacuous chart the same J -bound must beat a strictly positive self-energy

and sometimes loses. The selected sequence $a_k = 2^k$ is a negative census on the computed anchors, with no k_0 and no source-pure rate. No RH claim.